A Label-Free Runtime Safety Monitor for Frozen End-to-End Driving Planners,
Evaluated Closed-Loop at Benchmark Scale

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Abstract

End-to-end driving planners fail catastrophically in safety-critical closed-loop tests: published NeuroNCAP results score UniAD at 1.84 out of 5, with collisions in 88–98% of runs. These collision failures are invisible to the open-loop displacement-error metrics that dominate the field. We build a runtime monitor that reads only the frozen planner's own outputs: its planned trajectory, the objects it detects, and their tracked motion. The monitor uses no labels, no training, and no privileged simulator state. When it predicts a collision, it commands a full stop, holds that stop, and returns control only after consecutive clear frames confirm the threat has passed.

Across nineteen pre-registered campaign iterations and an independent verification pass we show six results. (1) The published NeuroNCAP UniAD baseline reproduces: rerunning the full benchmark at 20 runs per scenario yields a pooled score of 2.12 against the published 1.84. To our knowledge, we are the first to independently reproduce this baseline; a concurrent rerun by the DMAD authors (2.11), discussed in related work, corroborates the value. (2) The monitor, a union of two label-free geometric detectors with the threat-cleared stop-and-release policy, raises the full-benchmark score to **2.91** ($+0.783$, 95% CI $[+0.605, +0.928]$) while leaving behaviour in clean scenes, scenes with no staged threat, identical to the unmonitored planner. Its cost on our deployment metric, which combines safety with driving progress, is statistically indistinguishable from zero (-0.03, 95% CI $[-0.13, +0.07]$). (3) No softer intervention we tested could replace the committed stop. Three evasive maneuvers, a calibrated crawl, and a geometric threat-router all failed their pre-registered falsification criteria. The router failed narrowly, on a single misrouted crossing, while showing that a monitor with a positive deployment effect is achievable ($+0.226$ vs. the unmonitored planner, CI $[+0.004, +0.421]$). (4) An RSS-style formal safety envelope on identical inputs achieves the campaign's best raw safety scores, but only by nearly refusing to drive. This quantifies in closed loop the over-conservatism of formal envelopes that the literature so far only asserts. (5) We locate why the planner cannot save itself. In safety-critical frames, where a collision is imminent, the planner's candidate trajectories collapse onto its dangerous main plan: UniAD's candidates span 14 m of diversity on benign frames (frames where the planner is not on a collision course) but only 4 cm under imminent collision, and VAD retains partial diversity that stays below a pre-registered viability bar. A separately trained candidate generator that reads the frozen planner's internal planning embeddings is accurate on benign frames, yet on the same dangerous frames it produces **zero feasible escapes**, where a feasible escape is an alternative trajectory that is both kinematically drivable and collision-free. The collapse therefore sits in the planner's internal planning representation, not in any decoder above it. (6) Two independent failure analyses, one on cross-planner transfer and one on intervention routing, converge on the same bottleneck: the quality of the object tracking the monitor consumes.

One of our own earlier headline claims, that the monitor's deployment effect was net positive, was withdrawn by our own audit after we found that deterministic episode replays had been pooled as if they were independent samples; the claim was later re-established on genuinely independent episodes. Six experiments whose pre-registered hypotheses failed are published in full rather than discarded. All raw per-frame evidence is already released: it is committed in the public repository, and every number in this paper regenerates from it.

1 Introduction

Open-loop planning metrics on nuScenes are saturated and gameable: an ego-state MLP with no perception outperforms published end-to-end planners on displacement error [2]. Closed-loop evaluation under safety-critical perturbation tells the opposite story: state-of-the-art planners collide in the vast majority of adversarial episodes [1]. Runtime assurance belongs in this gap. If a frozen planner's failures can be predicted from its own outputs, a small monitor can intervene before they become collisions, without retraining, fleet data, or privileged simulator state.

This paper reports a complete measurement campaign on that thesis. Its contributions are the six numbered results in the abstract. Its method matters as much as its numbers: every iteration of the campaign was pre-registered, with falsification criteria written down before any data was collected; every comparison pairs the monitored and unmonitored arms on identical episodes; every number can be regenerated from raw evidence committed in the public repository; and the campaign's one statistical over-claim was caught by our own audit, withdrawn, and re-established on fresh data.

2 Related work

Runtime monitors for end-to-end planners. CATPlan/RiskMonitor [6] is the canonical learned monitor: a transformer decoder trained on the sign of the planner's collision loss, reading UniAD/VAD internal queries, paired with a braking policy (a 66.5% closed-loop collision-rate reduction on NeuroNCAP). Argus [7] attaches a takeover fallback that generates its own replacement trajectory in CARLA. Centaur [8] adapts the planner's weights at test time. Our monitor differs on each axis. It uses label-free geometry over the planner's own outputs, with no trained head. It never invents a replacement trajectory; the failures of our evasion, crawl, and routing experiments (§6) suggest this restraint is not merely simpler but structurally safer. It performs no weight updates. And it is validated with a progress-aware metric that none of the above report.

Retrained planners on the same benchmark. Concurrent work improves NeuroNCAP scores by retraining or replacing the planner rather than monitoring it: BridgeAD [3] reports 2.98–3.06 under the same trajectory post-processing protocol in which UniAD scores 1.84, and a VLM-agent system reports 3.49 in the protocol without post-processing [5]. Those results are complementary rather than comparable to ours: they change the planner, while this campaign measures what a training-free monitor buys on the planner exactly as shipped. Independently, DMAD's own rerun of the UniAD baseline (pooled 2.11) [4] corroborates the reproduction reported here (2.12).

Runtime selection among a frozen planner's candidates. Scoring and re-ranking a planner's candidate trajectories is an active paradigm [9, 10, 11], but in every instance we could verify, the scorer is co-trained within the planner's own pipeline and evaluated on the non-reactive NAVSIM benchmark. We found no published work that re-ranks a *frozen* planner's native candidates at runtime using an external safety signal. Our measurements close this mechanism for the planner's command-indexed candidates, which collapse precisely when they are needed (§7), and reopen it for separately trained candidate generators.

Mode collapse under threat is documented as a training-time phenomenon [12], with explicit metrics reported only for trajectory prediction [13]. To our knowledge, our measurements of candidate diversity conditioned on imminent collision, taken closed-loop on two planners' own candidate sets, are the first of their kind. Likewise, we found no prior closed-loop report of deployment-aware statistics for a runtime monitor (combined progress-and-safety scores with bootstrap confidence intervals, intervention budgets, detection lead time), and no prior closed-loop quantification of how over-conservative a formal safety envelope [14] is in practice.

3 Apparatus and methodology

The apparatus is three public containers on one L4 GPU: the NeuroNCAP orchestrator, which drives the scenario actor and scores each episode; the NeuRAD neural renderer, which produces photoreal multi-camera frames from real nuScenes drives; and the frozen planner, which serves `/infer` and carries the monitor as a ∼150-line patch that an environment variable enables or disables per experimental arm.

NeuroNCAP episodes replay deterministically for a given run index. We established this property mid-campaign. It first invalidated an early pooled claim (§9), and it then strengthened the design: because both arms replay the identical episode at the same run index, every comparison in this paper is a paired comparison on identical episodes. We call such comparisons *seed-paired*. The determinism is verified at scale: the first six run indices of the 20-run measurement reproduce the earlier 6-run measurement exactly, on every scenario pair, in both arms.

The same methodology held for all nineteen iterations of the campaign. Hypotheses and falsification criteria were frozen in writing before data collection. Every reported delta carries a seed-paired bootstrap confidence interval. Per-frame monitor decision logs, per-run trajectories, and run logs are committed for every arm. Experiments whose pre-registered hypotheses failed are published with the same weight as those that succeeded.

4 The monitor

The monitor brakes when either of two conditions holds: the closest approach between the planner's intended path and a tracked object's path falls below 1.5 m, or the observed time-to-collision with a closing agent falls below 2.5 s. The two conditions catch physically distinct failure modes. The first catches side crossings, where the planned path and an object's path genuinely intersect. The second catches the head-on approach that the planner's own optimism hides: on runs that end in collision, the planner's plan claims 3–4 m of clearance. Object velocity is *observed*, by ego-motion-compensated tracking of each object under a persistent identity, rather than taken from the planner's motion forecast; the forecast is optimistic on exactly the runs that collide. The underlying signal is real and cheap: the planner's own outputs predict its collisions with an AUROC of 0.83.

When the monitor triggers, the stop is *latched*: the vehicle stays stopped even if the trigger was spurious, which makes a false alarm safe. The latch *releases*, returning control to the planner, only after four consecutive frames verify that the planner's current plan is clear of threats. We call this full configuration, the two detectors plus the threat-cleared release, the *released union*. Throughout the paper, *safe-progress* denotes our deployment metric: the product of episode safety and driving progress. The release strictly dominates a latch that never releases: safety is identical, and safe-progress improves by +0.246 (CI [+0.206, +0.293]).

pooled, 14 official scenes (n=20/pair)	unmonitored UniAD	+ released union
NeuroNCAP score (0–5)	2.12 (published 1.84)	**2.91**
side collisions	74%	**44%**
stationary collisions	29%	**18%**
frontal (score / collisions)	1.24 / 78%	1.78 / 90%
safe-progress	2.40	2.36

Table 1: The full-benchmark measurement: 799 episodes, seed-paired. Benchmark delta +0.783, CI [+0.605, +0.928]; safe-progress delta −0.03, CI [−0.13, +0.07].

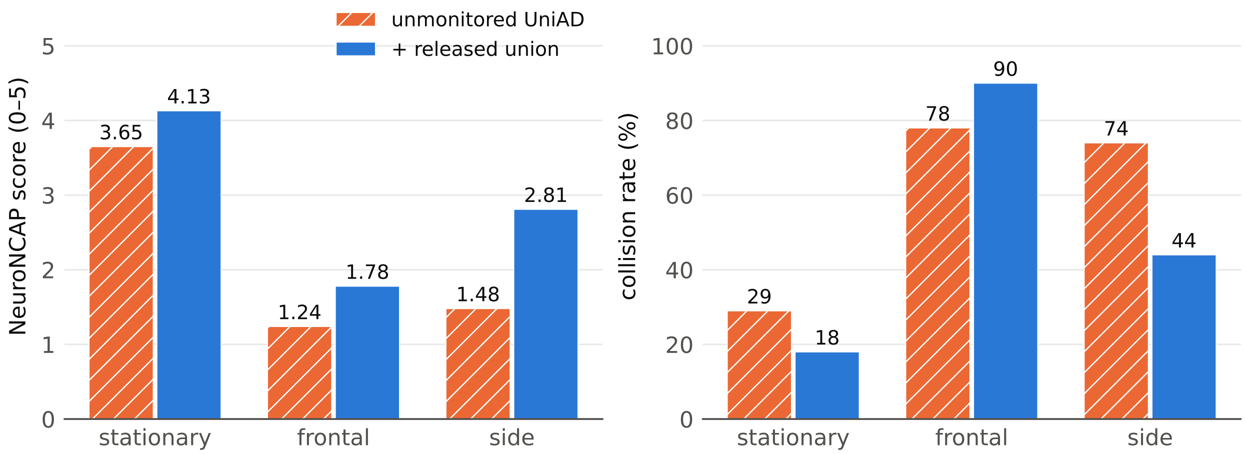

Figure 1: Per-class results at n=20 runs per pair. In frontal scenarios the monitor mitigates collisions rather than preventing them; the regression on the frontal 0346 scenario is confirmed real and is reported as a named cost of the stop policy.

5 The benchmark result

In the units a safety case uses, all computed from the committed decision logs: the median detection lead is 2.5 s before the moment of counterfactual contact, that is, before the collision that occurs in the unmonitored arm (Fig. 2); the monitor spends 11 brake frames per 242 benign meters driven; and the mean frontal impact speed drops from 13.9 to 6.7 m/s.

6 Negative results I: the stop is a position guarantee

Five designs that aimed to preserve more driving progress than the full stop were pre-registered and refuted. The pattern of their failures is itself the finding.

Three evasive maneuvers fail under *false* alarms: an evasive action taken on a spurious trigger creates a crash where none was coming, and the third design crashes the benign scene in 25% of runs against the unmonitored planner's 10%. A calibrated 2.0 m/s crawl fails under *true* alarms: instead of halting short of a crossing object's path, as the stop does, it delivers the ego vehicle into the crossing point, raising side collisions from 37% to 57%. A geometric threat-router, which commands a stop when the projected object path overlaps the planned corridor and a crawl otherwise (Fig. 3), comes closest. It records the campaign's first deployment-metric confidence interval excluding zero against the unmonitored planner (+0.226, CI [+0.004, +0.421]), yet it still fails its pre-registered

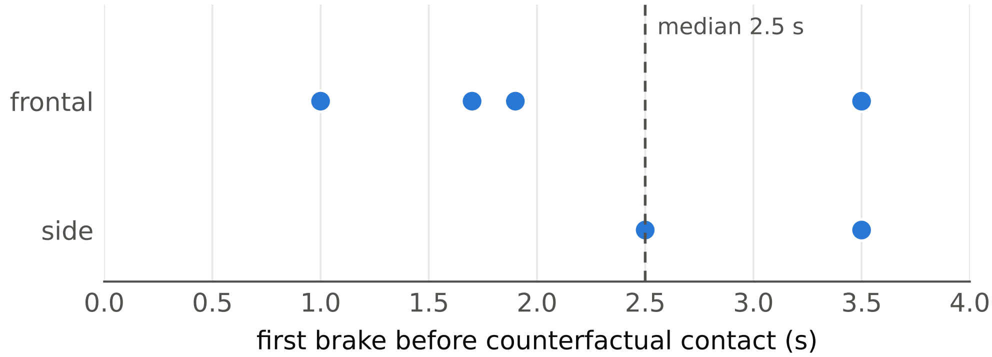

Figure 2: Detection lead time: the six lead-time events measurable under the strict contact criterion, shown as individual events because six values are too few for a histogram.

safety gate on a single misrouted crossing.

The conclusion is twofold. First, the stop is not merely a speed reduction but a *position guarantee*: a stopped vehicle cannot be delivered into a collision point, so the stop is safe whether the alarm is true or false, and it was the only tested intervention with that property. Second, the router shows that a deployment-positive monitor is demonstrably achievable, pending a routing signal that is safe on crossings (§8).

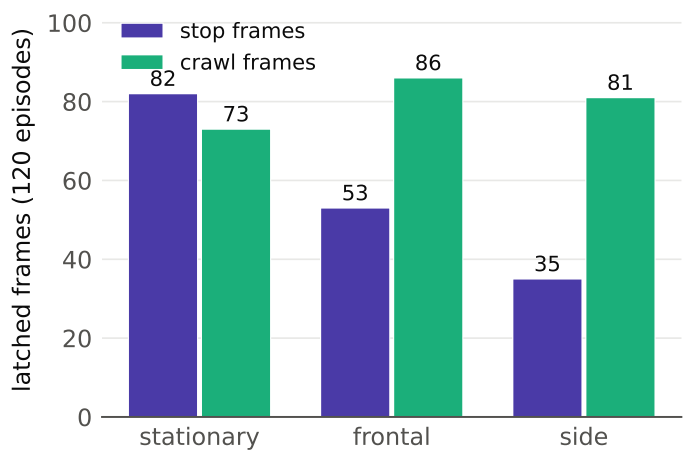

Figure 3: Frames in which the threat-router intervened, split by chosen action and scenario class. The crawl frames account for the router's deployment gain; the misrouted side-class crossing accounts for its safety failure.

7 Negative results II: no plan B, envelope paralysis

The planner has no plan B. The natural escape from the stop's progress ceiling is to execute a safer trajectory that the planner itself already proposes; such a takeover is safe on false alarms by construction. Pre-registered checkpoints therefore measured whether a low-risk candidate exists at the moments the executed plan is dangerous. It does not. UniAD's command-conditioned candidates span 13.9 m of diversity on benign frames and collapse to a 4 cm spread under imminent collision (0 escapes in 37 dangerous frames). VAD's native modes retain partial diversity (escapes in 21% of those frames) but stay below the frozen 30% viability bar. We then trained the motivated successor: a 1.2M-parameter candidate head that generates K=8 trajectories with an explicit diversity objective, conditioned on the frozen planner's own planning-query embeddings and trained on 60 scenes disjoint from all evaluation. The head is a faithful generator: on benign frames its best-of-K error stays within its pre-registered fidelity bar. Its repulsion term forces its candidates to diverge under threat. And on the same 37 dangerous frames it yields **0 feasible escapes**: 16 diverging candidates, every one violating kinematic limits. Three measurements by three routes agree. The deficit is a property of the planning representation itself: under imminent collision, that representation no longer contains a feasible alternative for any decoder to find.

Stopping power is free; selectivity is not. An RSS-style guaranteed-stopping envelope, given inputs and an actuator identical to the monitor's, posts the campaign's best raw safety (clean scenes 0% collisions, frontal 30%, side 0%). It does so by driving 3.6–8.2 m where the planner drives 21–32 m, for a safe-progress of 0.88 against the unmonitored planner's 1.83 (union minus RSS = +1.345, CI $[+0.944, +1.701]$).

8 The binding constraint is tracking quality

Two failure analyses, carried out from independent directions, converge on the same constraint.

Transfer. On a frozen VAD planner, the union prevents exactly the failures VAD has (stationary collisions fall from 85% to 0%, side collisions from 65% to 0%) but brakes far too often everywhere else. The decision logs attribute the over-braking to identity jitter: VAD exposes no learned tracker, so nearest-neighbor association switches object identities between frames, and each switch manufactures spurious closing speed.

Routing. The router's single misrouted crossing traces to velocity dropout across identity switches: when an object's identity changes, its velocity estimate momentarily vanishes, and the projected-path test misses the crossing. All three named per-frame geometric repairs were refuted offline on the committed logs: one is provably a no-op, one trades away the mechanism it was meant to save, and one relies on a feature that does not separate unsafe frames from safe ones.

In both analyses the discriminating signal is velocity continuity through identity switches, a property of the tracking layer rather than of any decision rule built on top of it. An association-and-filter layer built to this specification converts 12 of 13 unsafe crawl frames in offline replay, yet fails its pre-registered every-frame bar by a single borderline frame (0.2 m). The gate held, and the layer never ran closed-loop. That result, together with the caveat that offline replay may not perfectly reproduce closed-loop behaviour, is part of the released record.

9 Verification, reproducibility, and the incident record

An independent verification pass re-derived every claim from the raw evidence. It discovered the episode determinism described in §3, and it found that an early pooled validation had counted deterministic replays of the same episodes as independent replications. We withdrew that headline claim and re-measured on genuinely unique episodes, where the claim was re-established (mini-scene safe-progress +0.398, CI $[+0.133, +0.665]$).

The infrastructure incidents of the full-benchmark measurement are reported as part of the measurement. Five machine-freezing events occurred across two physical hosts; an on-box vitals instrument traced them to memory exhaustion on a swapless image. One scenario pair is reported at n=19 because its final episode reproducibly froze the pre-swap host. No completed episode was lost or re-measured differently; the exact reproduction of the earlier 6-run measurement is the proof. Every number in this paper regenerates from raw evidence committed in the public repository, using the commands in the repository README.

10 Limitations

This campaign uses one simulator (NeuroNCAP with the NeuRAD renderer), whose deterministic replay we characterize and exploit; 20 runs per scenario pair against the published 100-run protocol; and two planners. The VAD transfer result is bounded by the quality of the identity-association layer. The RSS baseline implements the longitudinal form only. There is no fleet data and no real-vehicle validation. Preventing frontal head-on collisions, as opposed to mitigating them, remains open, and on this evidence it is not reachable by maneuver invention from a label-free monitor.

11 Conclusion

A label-free monitor on a frozen planner lifts the full official NeuroNCAP benchmark from an independently reproduced 2.12 to 2.91, with a confidence interval excluding zero, at a deployment-metric cost statistically indistinguishable from zero. The campaign's negative results carry as much weight as its wins: the stop's position guarantee, the planner's missing plan B, the formal envelope's paralysis, the deployment-positive router that failed its safety gate, and the convergence of two independent failure analyses on tracking quality. Together they map the mechanism space for runtime driving assurance more sharply than the headline number does, and they name the next constraint to remove.

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